

Focal Region Shape and Energy Approximation for Frequency Diverse Array

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Abstract—This work investigates the analysis and design of frequency diverse arrays (FDAs), with a particular focus on focal region geometry and energy concentration. Although different frequency offset distributions are known to yield distinct FDA beampatterns, existing studies rely largely on simulations and offer limited analytical insight. We develop a Maclaurin-series-based approximation that characterizes the focal region geometry and provides a closed-form approximation of the in-threshold energy. Under identifiable conditions, the FDA exhibits an ellipse-like spotlight focal region; otherwise, the resulting beampattern is non-elliptical. The proposed approach is computationally efficient, shows strong agreement with simulation results, and offers clear analytical guidance for FDA design.

Index Terms—Frequency diverse array (FDA), beampattern analysis, focused beamforming, computationally efficient evaluation, inverse optimization

I. INTRODUCTION

Frequency diverse array (FDA) was first introduced in the paper [1]. It introduces small frequency offsets across adjacent elements creating an additional frequency offset phase term which enables the FDA's specific range-angle coupled beampattern. FDA can provide additional spatial degrees of freedom beyond conventional phased arrays, which provides it with a wider range of potential applications [2]–[4].

The evaluation and design of frequency offsets have been central topics in FDA research, as they directly shape the resulting beampattern. A uniform frequency offset produces an S-shaped FDA beampattern [1], [5], which has served as a canonical example for illustrating the inherent time variation and range-angle coupling effects of FDAs. These two characteristics underpin much of FDA research and define the fundamental challenges that frequency offset design aims to address.

The time variation of FDA can be undesirable in certain applications. Although combining FDA with multiple-input multiple-output (MIMO) radar provides a viable strategy [6], [7], a more fundamental and enduring approach is to design frequency offsets that achieve time-invariant FDA [8]–[11].

In addition, with the range-angle coupling property of the FDA, many studies have focused on frequency offsets which can steer the energy to a desired focal point, including classical nonuniform designs [12]–[14] and random offset schemes [15], [16]. Moreover, appropriate frequency offsets can also enable FDA focusing in the short-range regime [10], [17], [18]. These examples highlight the central role of frequency offset in shaping the FDA beampattern.

However, despite this progress, the design process still lacks quantitative guidelines, as general, quantitative, and physically interpretable criteria that directly evaluate frequency offset remain limited, which motivates the need for a principled evaluation framework to support frequency offset design. For instance, the standard deviation of the frequency increment (SDFI) in [18] offers a convenient metric. However, it mainly serves as a relative descriptor and it is not readily physically interpretable for constraining the optimization of frequency offset. And [19] shows for a terahertz (THz) linear array that the 3 dB focal region can be approximated by an ellipse in range-angle plane. However, that formula relies on the THz near-field assumption and dipole coefficients (X, Y, Z), making it cumbersome and un-intuitive, and it mainly covers only the 3 dB contour. Overall, existing methods are predominantly diagnostic rather than constructive, leaving a gap for a general evaluation framework that supports multi-threshold analysis and can directly inform frequency offset design.

To address the limited ability of existing methods to quantify focal region shape and energy and guide offset design, we propose an approximation framework for FDA focal region. At the geometric level, a Local Quadratic Correlation Approximation (LQCA) is derived from a second-order expansion of the steering vector correlation, whose coefficients are expressed only through the numerical features of the frequency offset and array index. LQCA offers an explicit, physically interpretable description of the focal region shape, predicting ellipse-like shape under identifiable conditions and revealing non-elliptical case when those conditions fail. At the energetic level, we further develop a focal region energy approximation based on LQCA. The LQCA enables a joint analysis of focal region geometry and energy, and offers simple but effective metrics for the evaluation and design of FDA frequency offset.

II. FDA MODEL

Consider a uniform linear array (ULA) FDA consisting of M elements with inter-element spacing d , as shown in Fig. 1. The signal transmitted by the m -th element in FDA is

$$s_m(t) = e^{j(2\pi f_m t + \varphi_m)}, \quad 0 \leq m \leq M-1, \quad t \in [0, T], \quad (1)$$

where φ_m is the phase shift, T is the duration of the pulse, and f_m is the frequency of the m -th element. The f_m is expressed as

$$f_m = f_0 + \Delta f_m, \quad (2)$$

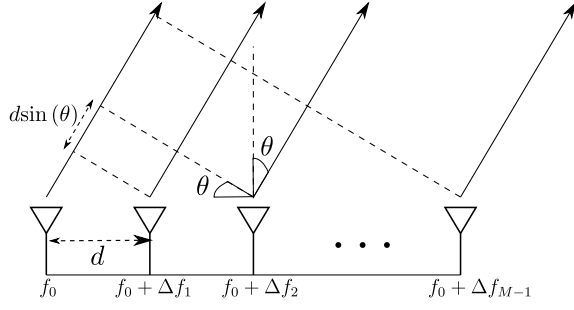


Fig. 1. Sketch of FDA with uniform linear array

where f_0 is the carrier frequency of the first element (reference element) and Δf_m is the frequency offset. The overall signal arriving at the observation position (R, θ) (R is the range in meters, θ is the azimuth angle in degrees, both with respect to the first array element) is

$$S(t, R, \theta) = \sum_{m=0}^{M-1} s_m(t - \frac{R_m}{c}) = \sum_{m=0}^{M-1} e^{j(2\pi f_m(t - \frac{R_m}{c}) + \varphi_m)}. \quad (3)$$

In the far-field case, $R_m \simeq R - md \sin \theta$. Considering only the phase shift of the weights and $\Delta f_m \ll f_0$, (3) can be expressed as

$$\begin{aligned} S(t, R, \theta) &= \sum_{m=0}^{M-1} e^{j(2\pi(f_0 + \Delta f_m)(t - \frac{R - md \sin \theta}{c}) + \varphi_m)} \\ &\approx e^{j2\pi f_0(t - \frac{R}{c})} \\ &\quad \sum_{m=0}^{M-1} e^{j(2\pi(\Delta f_m t - \frac{\Delta f_m R}{c} + m \frac{f_0 d \sin \theta}{c}) + \varphi_m)}. \end{aligned} \quad (4)$$

Eliminating terms unrelated to the array index, m , in (4), the array factor (AF) of FDA can be shown as

$$\begin{aligned} AF(t, R, \theta) &= \sum_{m=0}^{M-1} e^{j\phi_m} \\ &= \sum_{m=0}^{M-1} e^{j(2\pi(\Delta f_m(t - \frac{R}{c}) + \frac{2\pi f_0 m d \sin \theta}{c}) + \varphi_m)}. \end{aligned} \quad (5)$$

where ϕ_m is the total phase. For steering the FDA to beam-forming focus point (R_d, θ_d) , φ_m could be designed as:

$$\varphi_m = 2\pi \Delta f_m \frac{R_d}{c} - 2\pi f_0 \frac{md \sin \theta_d}{c}. \quad (6)$$

With phase shift shown in (6), the final AF of FDA is

$$AF(t, R, \theta) = \sum_{m=0}^{M-1} e^{j2\pi(\Delta f_m(t - \frac{R - R_d}{c}) + f_0 \frac{md(\sin \theta - \sin \theta_d)}{c})}. \quad (7)$$

III. FRAMEWORK DERIVATION

Please note that the terms “mean,” “variance,” and “covariance” used in this paper are employed solely as algebraic summaries of the deterministic sequences $\{\Delta f_m\}_{m=0}^{M-1}$ and $\{\phi_m\}_{m=0}^{M-1}$ for notational convenience; they do not imply any underlying probabilistic model or statistical inference.

A. Local Quadratic Correlation Approximation (LQCA)

We propose local quadratic correlation approximation (LQCA) to approximate the shape of the FDA's focal region.

Consider a fixed time t . Let δR and $\delta \theta$ denote small perturbations around (R_d, θ_d) . Define $s \triangleq \sin \theta$. Hence, $\delta s = \sin(\theta_d + \delta \theta) - \sin \theta_d$. The bias of the phase $\delta \phi_m$ at each element is

$$\begin{aligned} \delta \phi_m &= \phi_m(R_d + \delta R, \theta_d + \delta \theta) - \phi_m(R_d, \theta_d) \\ &= -2\pi \Delta f_m \frac{\delta R}{c} + 2\pi \gamma m \delta s, \end{aligned} \quad (8)$$

where $\gamma \triangleq \frac{f_0 d}{c}$ is a constant and c is wave speed.

With the bias phase $\delta \phi_m$, the bias steering vector is $\mathbf{a}(\delta R, \delta s) = [e^{j\delta \phi_0}, e^{j\delta \phi_1}, \dots, e^{j\delta \phi_{M-1}}]^T$ and the zero-bias (i.e., at point (R_d, θ_d)) steering vector is $\mathbf{a}(\delta R = 0, \delta s = 0) = [1, 1, \dots, 1]^T$. Apply normalized inner product to these two vectors to derive their correlation A

$$A(\delta R, \delta s) = \frac{1}{M} \mathbf{a}(0, 0)^H \mathbf{a}(\delta R, \delta s) = \frac{1}{M} \sum_{m=0}^{M-1} e^{j\delta \phi_m}. \quad (9)$$

Since δR and δs remain small over the entire space, the correlation A can be approximated using the Maclaurin expansion, $e^{jx} \approx 1 + jx - \frac{1}{2}x^2$. Correlation A is rewritten as

$$\begin{aligned} A(\delta R, \delta s) &\approx \frac{1}{M} \sum_{m=0}^{M-1} (1 + j\delta \phi_m - \frac{1}{2}\delta \phi_m^2) \\ &= 1 + j \frac{1}{M} \sum_{m=0}^{M-1} \delta \phi_m - \frac{1}{2} \frac{1}{M} \sum_{m=0}^{M-1} (\delta \phi_m)^2 \\ &= 1 + j\mu_1 - \frac{1}{2}\mu_2, \end{aligned} \quad (10)$$

where $\mu_1 = \frac{1}{M} \sum_{m=0}^{M-1} \delta \phi_m$ is the mean of the phase shift $\delta \phi_m$ and $\mu_2 = \frac{1}{M} \sum_{m=0}^{M-1} (\delta \phi_m)^2$.

According to the equation between variance and mean $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, the variance of $\delta \phi_m$ is expressed as

$$\text{Var}(\delta \phi_m) = \mu_2 - \mu_1^2. \quad (11)$$

To align with the pattern—which captures the array's power response, we calculate the power of the correlation A

$$\begin{aligned} |A(\delta R, \delta s)|^2 &\approx (1 + j\mu_1 - \frac{1}{2}\mu_2)(1 - j\mu_1 - \frac{1}{2}\mu_2) \\ &= 1 - (\mu_2 - \mu_1^2) + \frac{1}{4}\mu_2^2. \end{aligned} \quad (12)$$

For small $\delta \phi_m$, μ_1^2 and μ_2 are $O(\delta \phi_m^2)$, while μ_2^2 is $O(\delta \phi_m^4)$. Therefore, the μ_2^2 term can be neglected, and combining (11) with (12) gives

$$|A(\delta R, \delta s)|^2 \approx 1 - (\mu_2 - \mu_1^2) = 1 - \text{Var}(\delta \phi_m). \quad (13)$$

Substituting the expression of $\delta \phi_m$ in (8) into (13) yields

$$\mathbf{z}^T \mathbf{Q}_R \mathbf{z} \approx k, \quad (14)$$

where

$$\mathbf{z} = \begin{bmatrix} \delta R \\ \delta s \end{bmatrix}, \quad k = \frac{1 - |A(\delta R, \delta s)|^2}{4\pi^2} = \frac{1 - |A(\mathbf{z})|^2}{4\pi^2},$$

$$\mathbf{Q}_{Rs} = \begin{bmatrix} \frac{\text{Var}(\Delta f_m)}{c^2} & -\frac{\text{Cov}(\Delta f_m, m) \gamma}{c} \\ -\frac{\text{Cov}(\Delta f_m, m) \gamma}{c} & \text{Var}(m) \gamma^2 \end{bmatrix}, \quad (15)$$

where $\text{Var}(\Delta f_m)$ is the variance of the frequency offset Δf_m sequence, $\text{Var}(m)$ is the variance of the array index m sequence and $\text{Cov}(\Delta f_m, m)$ is the covariance of Δf_m and m . $\text{Var}(m)$ can be calculated using

$$\text{Var}(m) = \frac{1}{M} \sum_{m=0}^{M-1} \left(m - \frac{M-1}{2} \right)^2 = \frac{M^2 - 1}{12}. \quad (16)$$

Hence, (14) correspond to conic sections. Since $|A| < 1$, then $k > 0$. Note that the degenerate case $|A| = 1$ is excluded, since the focal region contracts to a point. Therefore, when $\mathbf{Q}_{Rs}$ is positive definite ($\mathbf{Q}_{Rs} \succ 0$), (14) is an ellipse centered at the origin.

Observe that $\mathbf{Q}_{Rs}$ can be interpreted as the covariance matrix of the vector $[-\Delta f_m/c, \gamma m]^\top$, which indicate that $\mathbf{Q}_{Rs}$ is at least symmetric and positive semi-definite, $\mathbf{Q}_{Rs} \succeq 0$. Under typical FDA operation, $\text{Var}(\Delta f_m) > 0$, $\text{Var}(m) > 0$, and $\gamma \neq 0$.

Then there are two case: (i) **Ellipse focal region**. If Δf_m and m are not perfectly correlated, $|\text{Cov}(\Delta f_m, m)| < \sqrt{\text{Var}(\Delta f_m) \text{Var}(m)}$, then $\det(\mathbf{Q}_{Rs}) > 0$ and thus $\mathbf{Q}_{Rs} \succ 0$. In this case, the focal region exists, and its shape in the R - $\sin \theta$ plane can be approximated as an ellipse by (14); (ii) **Non-elliptical beampattern**. Otherwise, $\det(\mathbf{Q}_{Rs}) \leq 0$. Then $\mathbf{Q}_{Rs}$ is not positive definite and the FDA tested does not have the focal region. As a reference, for uniform offsets $\Delta f_m = m\Delta f$, we have $\text{Var}(\Delta f_m) = (\Delta f)^2 \text{Var}(m)$ and $\text{Cov}(\Delta f_m, m) = \Delta f \text{Var}(m)$, which yields $\det(\mathbf{Q}_{Rs}) = 0$. Therefore $\mathbf{Q}_{Rs}$ is only positive semi-definite and no ellipse-like focal region exists.

Note that the correlation A is essentially the normalized array factor AF . Therefore, A and AF are related by

$$AF(\mathbf{z}) = MA(\mathbf{z}), \quad |AF(\mathbf{z})|^2 = M^2 |A(\mathbf{z})|^2. \quad (17)$$

B. Focal Region Energy Approximation Based on LQCA

Based on the ellipse approximation of the focal region shape provided by LQCA, we can evaluate the focal region energy within the prescribed normalized power threshold.

Expand k in (14) to rewritten the correlation A

$$|A(\mathbf{z})|^2 \approx 1 - 4\pi^2 \mathbf{z}^\top \mathbf{Q}_{Rs} \mathbf{z}. \quad (18)$$

To facilitate integration, AF is approximated by $e^x \approx 1 + x$

$$|AF(\mathbf{z})|^2 = M^2 |A(\mathbf{z})|^2 \approx M^2 e^{-4\pi^2 \mathbf{z}^\top \mathbf{Q}_{Rs} \mathbf{z}}. \quad (19)$$

For a prescribed normalized power threshold P_{th} , where $P_{th} = |A(\mathbf{z})|^2 \in (0, 1)$. Define $g \triangleq -\log P_{th}$. The threshold-matched contour region is an elliptic domain

$$4\pi^2 \mathbf{z}^\top \mathbf{Q}_{Rs} \mathbf{z} \leq g. \quad (20)$$

Accordingly, the focal region energy approximation based on LQCA, denoted by $P_{main}^{(2)}$, can be expressed as

$$P_{main}^{(2)} \approx \iint_{4\pi^2 \mathbf{z}^\top \mathbf{Q}_{Rs} \mathbf{z} \leq g} |AF(\mathbf{z})|^2 d\mathbf{z}. \quad (21)$$

Then whitening is applied to calculate $P_{main}^{(2)}$. Define $\mathbf{S} \triangleq 4\pi^2 \mathbf{Q}_{Rs}$ and $\mathbf{w} = \mathbf{S}^{1/2} \mathbf{z}$. Note that since $\mathbf{Q}_{Rs} \succ 0$, $\mathbf{S}^{1/2}$ exists. Then the exponent simplifies as $4\pi^2 \mathbf{z}^\top \mathbf{Q}_{Rs} \mathbf{z} = \mathbf{z}^\top \mathbf{S} \mathbf{z} = \|\mathbf{w}\|^2$ and the elliptic domain is mapped to a circular domain $\|\mathbf{w}\| \leq \sqrt{g}$.

With whitening $\mathbf{w} = \mathbf{S}^{1/2} \mathbf{z}$, the differential need to be transformed. Because $\mathbf{S} \triangleq 4\pi^2 \mathbf{Q}_{Rs}$ is a 2×2 matrix and $\mathbf{Q}_{Rs} \succ 0$, we can get $\det(\mathbf{S}^{1/2}) = \det(2\pi \mathbf{Q}_{Rs}^{1/2}) = (2\pi)^2 \det(\mathbf{Q}_{Rs}^{1/2}) = 4\pi^2 \sqrt{\det(\mathbf{Q}_{Rs})}$. Therefore, the differential is transformed to

$$d\mathbf{z} = |\det(\mathbf{S}^{-1/2})| d\mathbf{w} = \frac{1}{\det(\mathbf{S}^{1/2})} d\mathbf{w}$$

$$= \frac{1}{4\pi^2 \sqrt{\det(\mathbf{Q}_{Rs})}} d\mathbf{w}. \quad (22)$$

Therefore,

$$P_{main}^{(2)} \approx \frac{M^2}{4\pi^2 \sqrt{\det \mathbf{Q}_{Rs}}} \iint_{\|\mathbf{w}\| \leq \sqrt{g}} e^{-\|\mathbf{w}\|^2} d\mathbf{w}. \quad (23)$$

Using polar coordinates (ρ, ϕ) yields $\iint_{\|\mathbf{w}\| \leq \sqrt{g}} e^{-\|\mathbf{w}\|^2} d\mathbf{w} = \int_0^{2\pi} \int_0^{\sqrt{g}} e^{-\rho^2} \rho d\rho d\phi = \pi(1 - e^{-g}) = \pi(1 - P_{th})$. Hence,

$$P_{main}^{(2)} \approx \frac{M^2}{4\pi \sqrt{\det \mathbf{Q}_{Rs}}} (1 - P_{th}). \quad (24)$$

C. Feasibility Discussion on Higher-Order Corrections

Building on $P_{main}^{(2)}$, we examined two higher-order correction routes: (i) an omnidirectional higher-order cumulant expansion in the R - $\sin \theta$ plane, which makes the integral complex and difficult to analyze. (ii) an ellipse-preserving scheme that introduces fourth-order cumulants only along the two ellipse axes. The second route provides only marginal improvement and, in some cases, under-performs $P_{main}^{(2)}$. Considering both tractability and effectiveness, we conclude that higher-order corrections are unnecessary in our settings.

IV. EXPERIMENTAL RESULTS

This section aims to (i) evaluate the LQCA approximation's performance; (ii) compare the $P_{main}^{(2)}$ with beampattern sampling for focal region energy estimation.

In this study, all examples use an array of $M = 10$ elements with inter-element spacing $d = \frac{\lambda_0}{2}$. The initial frequency is set to $f_0 = 5$ GHz and the observation time to $t = 0$. The beam-forming focus point is located at $(R_d, \theta_d) = (500 \text{ km}, 50^\circ)$.

A. LQCA Performance

This section validates the proposed LQCA in two representative FDA: logarithmic [12] and Hamming window-based [14]. Their expressions are summarized in Table I.

For each case, reference contours are generated in the δR - $\delta \sin \theta$ plane at multiple power thresholds P_{th} and

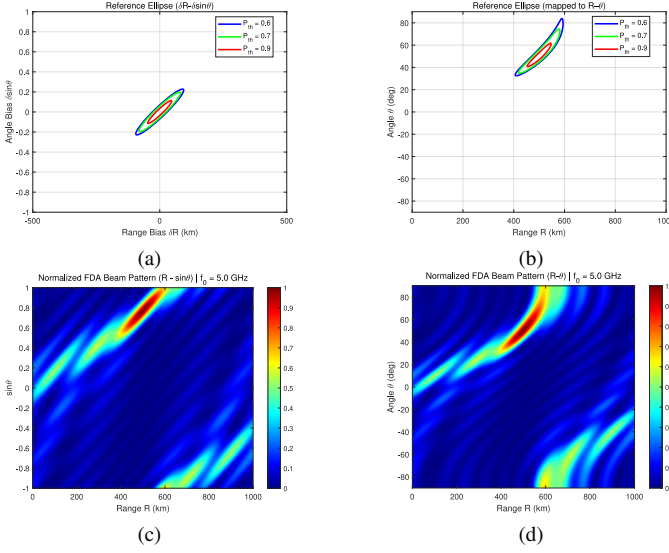


Fig. 2. LQCA validation with logarithmic frequency offsets: (a) reference ellipses in the $\delta R - \delta \sin \theta$ plane for three thresholds, (b) mapped contours in the $R - \theta$ plane. (c) and (d) normalized FDA beampatterns in $R - \sin \theta$ and $R - \theta$, respectively.

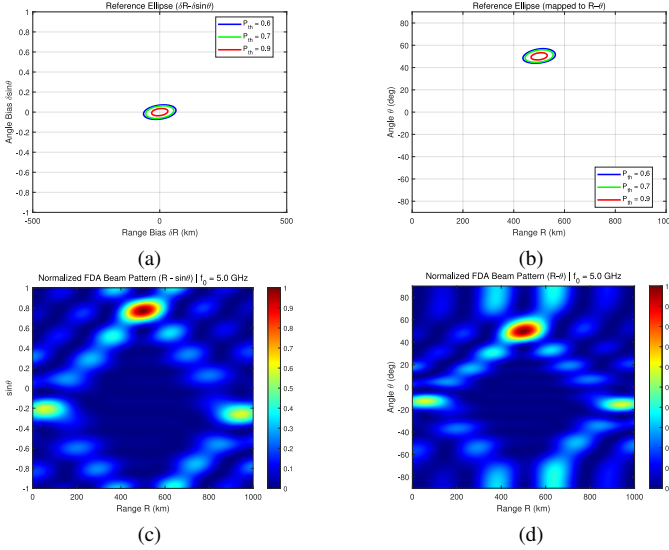


Fig. 3. LQCA validation with Hamming window-based frequency offsets: (a) reference ellipses in the $\delta R - \delta \sin \theta$ plane for three thresholds, (b) mapped contours in the $R - \theta$ plane. (c) and (d) normalized FDA beampatterns in $R - \sin \theta$ and $R - \theta$, respectively.

then mapped to the conventional $R - \theta$ plane via $(R_d + \delta R, \arcsin(\sin \theta_d + \delta s))$ for direct comparison with focal region shape in the normalized beampattern on the $R - \theta$ plane.

Fig. 2 shows the performance of LQCA in evaluating logarithmic frequency offset. In $\delta R - \delta \sin \theta$ plane (Fig. 2(a)), The predicted contours form a well-behaved family of ellipses. As the P_{th} increases, the contours contract toward the center while maintaining a stable elliptical axis direction. With mapping to $R - \theta$ plane Fig. 2(b), the reference shape aligns closely with the crescent-like shape of the focal region shown in

TABLE I
FREQUENCY OFFSETS FOR THE EXPERIMENT

Logarithmic FDA	$1.5 \log(m+1) \cdot 10^3$
Hamming Window-Based FDA	$1.5 \{0.54 - 0.46 \cos(\frac{2\pi m}{M})\} \cdot 10^3$

TABLE II
 $P_{main}^{(2)}$ VS. SAMPLED FOCAL-REGION ENERGY UNDER THREE THRESHOLDS (IN THE $R - \sin \theta$ PLANE)

Logarithmic FDA			
P_{th}	$P_{main}^{(2)}$	$P_{sampled}$	Rel. Err. (%)
0.6	2.0754×10^6	2.1229×10^6	2.238
0.7	1.5565×10^6	1.5708×10^6	0.907
0.9	5.1885×10^5	5.1915×10^5	0.058
Hamming Window-Based FDA			
P_{th}	$P_{main}^{(2)}$	$P_{sampled}$	Rel. Err. (%)
0.6	1.4061×10^6	1.3632×10^6	3.145
0.7	1.0546×10^6	1.0307×10^6	2.320
0.9	3.5153×10^5	3.4899×10^5	0.727

Fig. 2(d). The bending in Fig. 2(d) is consistent with the nonlinearity of the arcsin function, which provides an effective explanation for the crescent-shaped focal regions observed in focused FDA beampatterns (e.g., logarithmic and exponential frequency offsets). Comparing Fig. 2(a),(b) and Fig. 2(c),(d), the contours closely describe the focal region boundary, and their contraction rate with threshold aligns with the simulated focal region convergence.

Fig. 3 shows the results in Hamming window-based case, which further confirms the effectiveness of LQCA. The contractions observed in Fig. 3(a),(b) align with the convergence trend shown in the normalized beampatterns (Fig. 3(c),(d)). These observations serve as corroborative evidence that LQCA provides stable and visually accurate approximation across different frequency offset.

Across both logarithmic and Hamming window-based frequency offsets, LQCA provides contours that visually match the simulated focal region. In the $R - \sin \theta$ plane, the contours form stable ellipses. When mapped to $R - \theta$, the contours track the focal region ridge and naturally explain the crescent-like appearance. These results support LQCA as an effective method for evaluation and design of FDA.

B. Focal Region Energy Approximation

Table II compares the focal energy approximation from $P_{main}^{(2)}$ with the sampled simulation result. The relative error is defined as $Rel. Err. = \frac{|P_{main}^{(2)} - P_{sampled}|}{P_{sampled}} \times 100\%$.

Table II shows that, in the $R - \sin \theta$ plane, $P_{main}^{(2)}$ closely matches the sampled result. For logarithmic FDA, the relative errors at $P_{th} = 0.6/0.7/0.9$ are 2.238%/0.907%/0.058%; for Hamming FDA they are 3.145%/2.320%/0.727%. The error decreases as the threshold increases, indicating that $P_{main}^{(2)}$ accurately approximates the focal region energy with stable convergence and consistent performance across cases. Overall, $P_{main}^{(2)}$ provides an accurate, low-cost estimator for rapid pre-evaluation and design screening.

REFERENCES

- [1] P. Antonik, M. Wicks, H. Griffiths, and C. Baker, "Frequency diverse array radars," in *2006 IEEE Conference on Radar*, 2006, pp. 215–217.
- [2] H. So, M. Amin, S. Blunt, F. Gini, and W.-Q. Wang, "Introduction to the special issue on time/frequency modulated array signal processing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 11, no. 2, pp. 225–227, 2017.
- [3] J. Xu, G. Liao, S. Zhu, L. Huang, and H. C. So, "Joint range and angle estimation using mimo radar with frequency diverse array," *IEEE Transactions on Signal Processing*, vol. 63, no. 13, pp. 3396–3410, 2015.
- [4] H. Chen, W.-Q. Wang, Y.-S. Cui, and Y. Liao, "Statistical analysis for time modulation-based frequency diverse array beampattern," *IEEE Access*, vol. 7, pp. 84 142–84 154, 2019.
- [5] P. F. Sammartino, C. J. Baker, and H. D. Griffiths, "Frequency diverse mimo techniques for radar," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 49, no. 1, pp. 201–222, 2013.
- [6] J. Xu, J. Kang, G. Liao, and H. C. So, "Mainlobe deceptive jammer suppression with fda-mimo radar," in *2018 IEEE 10th Sensor Array and Multichannel Signal Processing Workshop (SAM)*, 2018, pp. 504–508.
- [7] L. Lan, G. Liao, J. Xu, Y. Zhang, and F. Fioranelli, "Suppression approach to main-beam deceptive jamming in fda-mimo radar using nonhomogeneous sample detection," *IEEE Access*, vol. 6, pp. 34 582–34 597, 2018.
- [8] Q. Cheng, J. Zhu, T. Xie, J. Luo, and Z. Xu, "Time-invariant angle-range dependent directional modulation based on time-modulated frequency diverse arrays," *IEEE Access*, vol. 5, pp. 26 279–26 290, 2017.
- [9] W. Khan and I. M. Qureshi, "Frequency diverse array radar with time-dependent frequency offset," *IEEE Antennas and Wireless Propagation Letters*, vol. 13, pp. 758–761, 2014.
- [10] A.-M. Yao, P. Rocca, W. Wu, A. Massa, and D.-G. Fang, "Synthesis of time-modulated frequency diverse arrays for short-range multifocusing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 11, no. 2, pp. 282–294, 2017.
- [11] A.-M. Yao, W. Wu, and D.-G. Fang, "Frequency diverse array antenna using time-modulated optimized frequency offset to obtain time-invariant spatial fine focusing beampattern," *IEEE Transactions on Antennas and Propagation*, vol. 64, no. 10, pp. 4434–4446, 2016.
- [12] W. Khan, I. M. Qureshi, and S. Saeed, "Frequency diverse array radar with logarithmically increasing frequency offset," *IEEE Antennas and Wireless Propagation Letters*, vol. 14, pp. 499–502, 2015.
- [13] L. Zhang, W. Xu, P. Huang, and W. Tan, "Comparison of frequency diverse array patterns with nonuniform frequency offset," in *2020 IEEE MTT-S International Wireless Symposium (IWS)*, 2020, pp. 1–3.
- [14] A. Basit, I. M. Qureshi, W. Khan, S. u. Rehman, and M. M. Khan, "Beam pattern synthesis for an fda radar with hamming window-based nonuniform frequency offset," *IEEE Antennas and Wireless Propagation Letters*, vol. 16, pp. 2283–2286, 2017.
- [15] J. Gao and X. Zhang, "Frequency diverse array antennas with random logarithmically increasing frequency offset," in *2023 5th International Conference on Intelligent Control, Measurement and Signal Processing (ICMSP)*, 2023, pp. 143–146.
- [16] Y. Liu, H. Ruan, L. Wang, and A. Nehorai, "The random frequency diverse array: A new antenna structure for uncoupled direction-range indication in active sensing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 11, no. 2, pp. 295–308, 2017.
- [17] Z. Wang, Y. Song, T. Mu, and Z. Ahmad, "A short-range range-angle dependent beampattern synthesis by frequency diverse array," *IEEE Access*, vol. 6, pp. 22 664–22 669, 2018.
- [18] W. Choi, A. Georgiadis, M. M. Tentzeris, and S. Kim, "Analysis of exponential frequency-diverse array for short-range beam-focusing technology," *IEEE Transactions on Antennas and Propagation*, vol. 71, no. 2, pp. 1437–1447, 2023.
- [19] L. Li, H. Li, Z. Chen, W. Chen, and S. Li, "An analytical range-angle dependent beam focusing model for terahertz linear antenna array," *IEEE Wireless Communications Letters*, vol. 11, no. 9, pp. 1870–1874, 2022.